

PORTABLE MODELS, RECONCILED BY COGNITIVE AGENTS

Ad hoc semantic reconciliation.

Can two systems that never agreed on anything build a shared understanding — and work together — on their own?

Cognitive agents, portable models, and where the human still decides

THE HOOK

Do two capable systems still need a data model agreed in advance to work together?

Or can they build a good-enough one **on their own**, for the job at hand?

*Decades of legacy systems that can't explain themselves, data of mixed quality, operations that must never break — all real.
The question is whether the pre-agreement, the slow and expensive part, is still required now that the systems can **reason**.*

the answer we reach: yes — with real qualifications

THE BURDEN

The standard-model quest never ends.

Everyone must agree one vocabulary in advance — by committee, before the systems ever meet.

It is expensive, slow, and permanently behind the world it models. Capable engineers build different but defensible models of the same network, and someone always has to reconcile them.

What if the systems could do that reconciling themselves?

THE REFRAME

A model that can stand on its own.

Once each side can **reason**, a system's data can be **lifted** into a self-describing form that any capable agent can pick up and use — **from the package alone, with nothing agreed in advance.**

Portable a sufficiently reasoning agent can make full use of it without asking anyone for clarification outside the package.

THE PATH

How do we get there — tractably, and safely?

This is **mapping a path**, not flipping a switch: inert legacy and cognitive systems coexisting, mixed-quality data handled gracefully, a light shared **anchor** making each step safe.

Can a lifted model really be understood cold?

When two models meet, can cognition bridge them with no pre-agreed standard?

A well-lifted model is understood cold — by strong, middle, and even weak readers.

Hand a lifted model to an independent agent that never saw the system; ask it to explain each concept from the package alone.



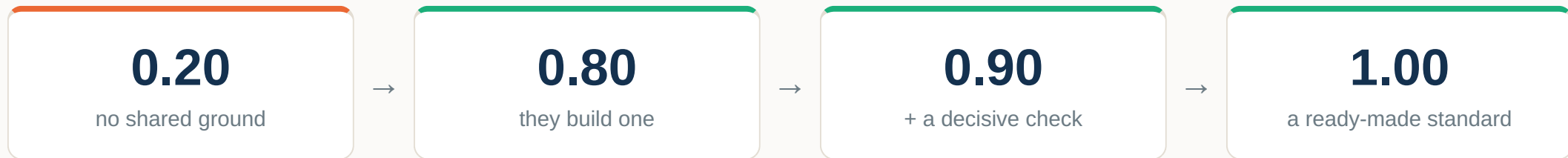
comprehension on the hardest cross-domain seam · perfect on routine material

Holds across six independent cases and across model families. Honest floor: the weak tier can confabulate — no magic.

When the ground isn't there, cognition builds it.

Two private models, no shared standard, nothing recognisable. Left alone, a careful agent refuses to guess and binds almost nothing.

Let the two agents **construct a thin shared reference themselves**, then bind through it:



They don't need a standard handed to them — they can make one.

A name with nothing behind it is a trap.

Portability is about **meaning**, not surface. A lift that keeps concept names but strips their meaning is **no safer than an anonymous one** — a bare name invites a plausible, wrong reading.

Same law for the shared anchor: a pointer with **no description is worse than nothing**. What makes a lift portable is meaning — and a reference used *in the lift* repairs a thin one.

THE TURN

The real prize: two systems that never agreed on anything, working together.

Portability is the object. **Reconciliation is the flagship use** — aligning two independently built models so the systems interoperate.

It is its own job, not a patch on a bad lift: two perfectly good models still diverge — one says *circuit*, the other *bearer*; a *grade* that looks shared but isn't — and bridging that divergence is the work.

They bind what matches — and refuse what only looks like it — with no standard.

Two agents reconcile two independently authored models **ad hoc**. The discipline is the point: a planted look-alike — a transport *grade* vs an IP *grade*, same word, unrelated meaning — is **refused**.

Where they need a nudge, the fix is a **thin shared glossary, not a universal standard** — and one shared anchor turns work that grew **quadratically** into work that grows **linearly** as systems are added.

Disciplined refusal, not naïve agreement.

Where both sides can reason, it finishes on its own — and it knows where to stop.

At the fully-cognitive end, reconciliation completes **autonomously** — negotiations and significance judgements included, no human in the loop. As cognition recedes it degrades honestly: it defers rather than guesses, and a thin reference buys back the facts an inert side can't supply.

⚠ The honest guardrail: a reference supplies information, never authority. The one decision that stays with a human is whose value governs when two sides genuinely conflict.

THE PROMISE HELD

Yes — with real qualifications.

- 1 Portability is real**
understood cold, across readers and model families
- 2 Cognition bridges divergence**
builds the ground, refuses the traps — no standard needed
- 3 It automates where both sides reason**
and hands off cleanly where authority, not meaning, is at stake

Every limit shown is a credibility signal, not a hedge.

THE CLOSE

This changes what we should standardise — and lightens the load.

Not “standards are over.” **Production and agreement can increasingly be ad hoc**, machine-to-machine, for the task at hand — coexisting with legacy, redirecting the standards effort toward the anchors and authority boundaries that actually need agreeing.

A roadmap, not a rupture.

The full evidence — six cases, four scenarios, cross-family checks — is in the report. Come see it.